KUNAL AGARWAL

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Education

Vellore Institute of Technology **Oct 2022 - Sep 2026**

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.08/10)

Experience

Sagacity Softwares Pvt. Ltd. **Feb 2025 - Apr 2025**

Software Developer Intern *Pune, Remote*

- Architected and shipped scalable RESTful APIs using Django and PostgreSQL, achieving 99% uptime and supporting reliable backend operations for production workloads.
- Optimized database schemas and query performance, improving API response time by 30% and enhancing system efficiency.
- Collaborated using Git-based workflows and code reviews, maintaining high code quality and adherence to industry-standard software engineering practices.

Projects

TheodoraQ | React.js, Node.js, Express.js, MongoDB, REST APIs, Vite, Material UI, OAuth 2.0 **Oct 2025 - Jan 2026**

- Engineered and deployed a production-ready AI-powered quiz platform (MERN stack) leveraging NLP-based prompt parsing, reducing manual quiz creation effort by 80% and supporting multi-role workflows.
- Secured end-to-end authentication and authorization (JWT, OAuth 2.0, RBAC) with real-time exam proctoring via tab-switch detection, preventing 90%+ of common cheating behaviors.
- Developed a REST API-driven analytics engine with 5+ interactive dashboards to track performance, accuracy, and topic mastery, reducing instructor evaluation time by 40% and enabling data-driven decision-making.
- Launched a live SaaS application (**theodoraq.dev**) with AI-assisted semantic search and real-time messaging, boosting student engagement by 2–3x and demonstrating production deployment, scalability, and system reliability.

MedVision AI | *Python, PyTorch, FastAPI, Streamlit, OpenCV, NumPy* **Mar 2026 - Apr 2026**

- Built an end-to-end AI-driven healthcare platform for Pneumonia (X-ray) and Brain Tumor (MRI) detection, training on 5,000+ medical images and achieving 93% average classification accuracy.
- Fine-tuned ResNet18-based deep learning models and optimized the inference pipeline to deliver real-time predictions with less than 1s latency on CPU.
- Integrated a Grad-CAM-based AI model that generated clinically relevant visual heatmaps, leading to a 30% increase in the speed of diagnosis interpretation for complex imaging studies in radiology departments.
- Orchestrated a production-ready FastAPI and Streamlit system supporting multi-model inference, automated PDF report generation, and modular architecture for future disease extensions.

IntelliFlow | *Python, FastAPI, PostgreSQL, Gemini API, JWT, Sentence Transformers, RAG* **Aug 2025 - Sep 2025**

- Spearheaded multi-tenant AI knowledge platform allowing 50 users to upload, organize, and retrieve documents, cutting average search time from 45 seconds to 20 seconds
- Implemented semantic search using 384-dimensional embeddings generated by Sentence Transformers and pgvector cosine similarity to retrieve relevant document passages based on user queries.
- Integrated Google Gemini API with a Retrieval-Augmented Generation (RAG) pipeline to generate context-aware answers from retrieved documents, including source references and conversation history.
- Developed a web-based chat interface with JWT authentication, tenant-level access control, and persistent conversation history, allowing users to create, revisit, and manage document-based conversations.

Technical Skills

Languages: Python, SQL
AI & ML: Ollama APIs, LLM Transformers, Semantic Search, RAG, PyTorch, TensorFlow, Scikit-learn, LangChain
Data Science: NumPy, Pandas, Data Preprocessing, Streamlit, Tableau
Backend & APIs: Django, Flask, RESTful APIs, FastAPI, JWT Authentication
Databases: MySQL, PostgreSQL, pgvector
Tools & Platforms: Git, GitHub, AWS, VS Code, Postman, Jupyter Notebook, Microsoft Excel, Power BI

Achievements & Certifications

Shell.ai Fuel Blend Challenge: Global Rank 46
Postman API Fundamentals Student Expert | CERTIFICATE
Data Analytics Job Simulation (Deloitte Australia, Forage) | CERTIFICATE
Artificial Intelligence Fundamentals (IBM SkillsBuild) | CERTIFICATE